

Environmental Drivers of Giant Virus Community Structure in Intertidal Mudflats

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Abstract

Despite the unique ecological significance of intertidal mudflats, the environmental parameters shaping giant virus community structures remain poorly understood. This study investigated 14 giant virus OTUs from the coast of China to systematically resolve how environmental filters govern viral distribution. PCoA revealed a dominant core cluster tightly associated with nutrient and carbon availability, whereas forward selection isolated seven key environmental predictors. dbRDA and Ward's hierarchical classification demonstrated geometric consistency, establishing an optimal biological partition at $k=4$. The identified clusters delineate a unique environmental factor, which demonstrates that environmental filtering actively structures giant virus biogeography in coastal intertidal ecosystems.

Introduction

Giant viruses, which refer to the viruses from the phylum Nucleocytoviricota, are known as ones that challenge the understanding of viruses due to their large size and genetic complexity. As the name suggests, giant viruses are characterised by their large size, comparable to that of bacteria and archaea (Aylward et al., 2021). The Nucleocytoviricota phylum comprises a monophyletic and heterogeneous group of viruses with double-stranded DNA, associated with infection across a wide range of hosts. The diversity of giant viruses has been increasingly identified across different ecological niches, including soil, water, and even mammals. Due to their unique characteristics and expanding diversity, many questions have been raised about their biogeographical distribution and the factors that dictate their survival in various environments (Ha et al., 2021).

Despite their ecological significance in supporting unique biodiversity and intricate food webs, the environmental parameters shaping giant virus community structure and distribution in intertidal mudflats remain poorly understood. This study addresses these knowledge gaps by investigating giant virus operational taxonomic units (OTUs) across diverse intertidal mudflat locations. Specifically, we aim to identify the key environmental factors driving giant virus communities through statistical selection, and model the direct relationships between specific viral taxonomic clusters and their corresponding environmental

constraints by coupling distance-based Redundancy Analysis (dbRDA) with hierarchical classification.

Materials and Methods

The dataset was derived from a survey of 199 intertidal mudflat metagenomes compiled by Ji et al. (2026), spanning 12 sampling locations along the coast of China (Figure 1a). To fit this project's scope, both the reconstructed giant virus OTUs and their corresponding environmental information were chosen for downstream community analyses. A total of 14 giant virus OTUs were identified, though the original authors did not elaborate extensively on these specific lineages. Raw data files and input matrices are available on a public GitHub repository (<https://github.com/mesosupp/NumEcolIndependentProject>).

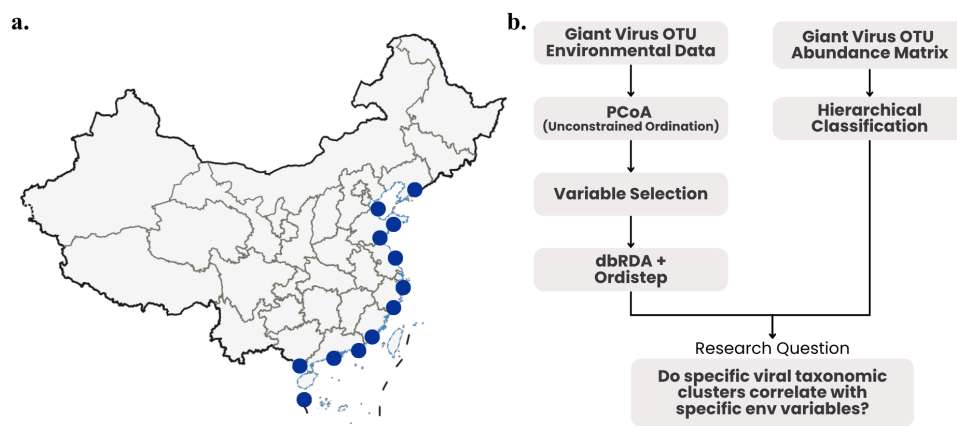


Figure 1. Geographic layout of intertidal sampling sites (a); computational pipeline for giant virus community analyses (b)

Following our analytical pipeline (Figure 1b), environmental matrices and OTU abundances were evaluated via parallel workflows. Environmental data underwent an unconstrained Principal Coordinates Analysis (PCoA) to map baseline community patterns, followed by ‘envfit’ vector fitting. To eliminate multi-collinearity, forward selection (ordiR2step) isolated the most significant predictors for a constrained distance-based Redundancy Analysis (dbRDA). Simultaneously, the OTU abundance matrix was classified hierarchically using Ward’s minimum variance method based on square-root transformed Bray-Curtis distances to resolve sample groupings across multiple cluster thresholds.

Results

To evaluate the ecological forces shaping giant virus communities, 11 environmental parameters were monitored across four functional groups: physical factors (Mean Annual Temperature (MAT), pH, and Moisture), salinity and dissolved ions (Salinity and Sulfate

(SO₄)), nutrient (Ammonium (NH₄), Nitrate (NO₃), Nitrite (NO₂), Total Nitrogen (TN), and Total Phosphorus, (TP)), and carbon pools (Total Organic Carbon (TOC)). Prior to multivariate modeling, moisture was excluded due to a large proportion of missing values. To resolve the baseline biogeographical structure of the giant virus OTUs, a PCoA based on Bray-Curtis distances was performed (Figure 2a). The first two axes captured a substantial portion of total community variance (PCoA1: 32.1%, PCoA2: 11.7%), revealing strong spatial clustering. A large core cluster comprising samples C, CXD, N, NB, S, SY, and W tightly aggregated on the right side of the ordination space.

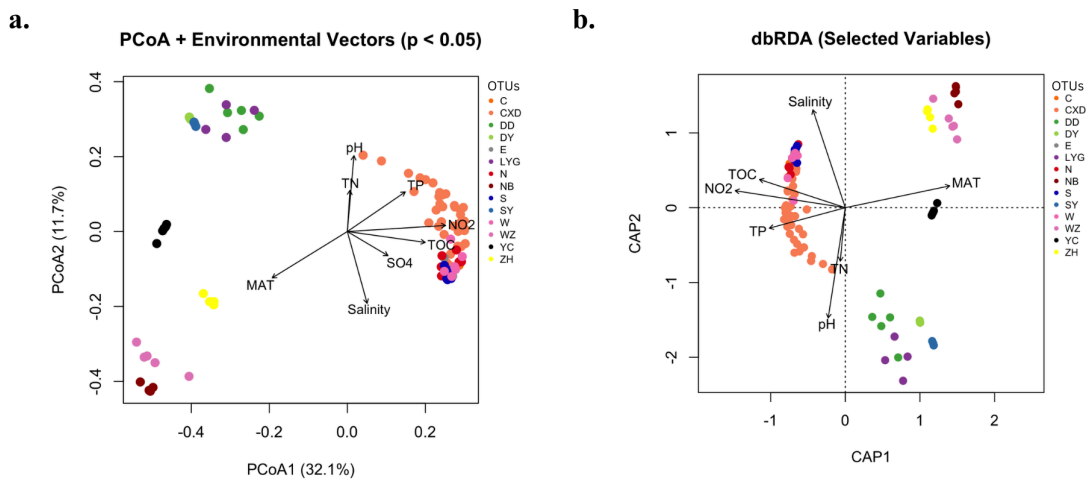


Figure 2. Multivariate ordination models of giant virus and environmental factors.

(a.) PCoA based on Bray-Curtis (b.) dbRDA with 7 forward-selected variables

Using the ‘envfit’ function, 1 non-significant variable ($p > 0.05$) was removed. The significant vectors revealed clear directional influences, where the massive core cluster aligned strongly with increasing gradients of NO, TP, and TOC, whereas outlying OTUs were driven by distinct physical pressures. Specifically, DD, DY, and LYG pulled toward the top-left quadrant without any of the variables, while others clustered separately, heavily influenced by MAT. To remove confounding multicollinearity and isolate the true ecological drivers, a forward selection procedure identified seven key environmental drivers, including MAT, Salinity, pH, NO₂, TN, TP, and TOC. A dbRDA was then performed to model the relationship between viral community structure and localized environmental niches (Figure 1b). The geometric configuration and biological groupings of the OTUs remained exceptionally consistent between the PCoA and dbRDA, despite an arbitrary mirroring of the horizontal axis where the dominant core cluster shifted to the left side.

To classify these samples into discrete groups, multiple hierarchical clustering configurations were validated (Figures 3a–f). Alternative linkage comparison shows highly disorganized, chained trees. Conversely, Ward’s method generated a clean, well-resolved

structure. A broad partition of $k=2$ simply separated the core estuary from regional outliers, a high-resolution partition of $k=4$ clusters matched the continuous ordination space perfectly. Cluster 1 aggregates the dense, nutrient-rich estuarine core group driven by TOC and TP. Cluster 2 isolates the hypersaline communities (DD, DY, and LYG) along the salinity gradient. Cluster 3 captures a narrow, specialized transition zone, and Cluster 4 groups the cold-temperature regional outliers driven by Mean Annual Temperature (MAT).

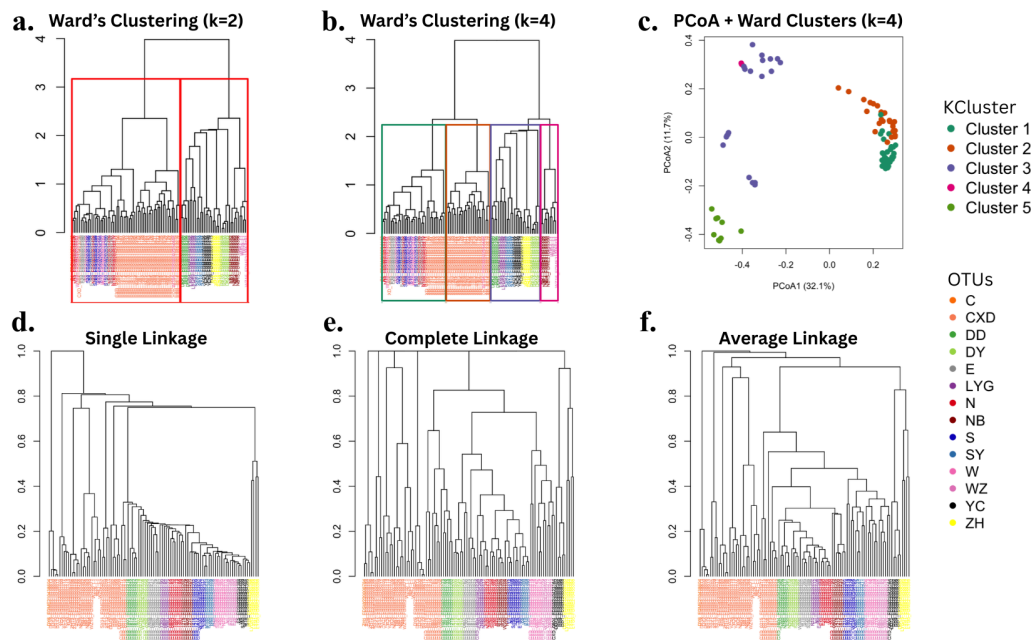


Figure 3. Structural validation of giant virus communities. (a) Ward's tree ($k=2$); (b) Ward's tree ($k=4$); (c) $k=4$ clusters overlaid onto the PCoA ordination; (d–f) alternative linkage comparisons (single, complete, average).

Discussion

The intertidal mudflat giant virus communities exhibit a stable biogeographical distribution shaped by deterministic environmental filtering. The strong geometric consistency between the PCoA, hierarchical clustering, and dbRDA proves these groupings represent genuine biological structures, not mathematical noise. The horizontal axis mirroring between the PCoA and dbRDA spaces is an arbitrary artifact of eigenvector solvers with zero biological significance; all relative sample distances and neighbor relationships remain perfectly intact. Because the constrained dbRDA replicates the unconstrained PCoA configuration so closely, the seven selected drivers clearly explain the community variance, confirming the "environment selects" paradigm in giant virus macroecology (Ji et al., 2026).

Choosing a hierarchical partition of $k=4$ clusters offers the optimal biological resolution for this dataset. A broad $k=2$ split completely masks localized environmental pressures, whereas over-clustering at $k=5$ artificially fragments the tightly packed estuarine

core group. The four-cluster layout highlights a prominent structural asymmetry: PCoA1 explains 32.1% of the total variance, nearly three times more than PCoA2 (11.7%), proving that the horizontal transition from stable, nutrient-dense conditions to physical stressors acts as the primary ecological sieve across China's coastlines.

The core-cluster in the multivariate ordination models reflects the niche breadths of the giant viruses and their cellular hosts. Cluster 1 represents a uniform, generalist viral community across multiple geographic sites (C, CXD, N, NB, S, SY, W), where high organic carbon (TOC) and nutrient availability (NO₂, TP) support stable, predictable host populations. Conversely, along the dominant horizontal axis, severe environmental filters trigger sharp community turnover. Cluster 2 isolates hypersaline-adapted communities (DD, DY, LYG), Cluster 4 isolates cold-temperature outliers driven by MAT, and Cluster 3 marks a narrow transitional zone. These distinct patterns prove that while generalist lineages dominate stable estuarine mudflats, localized salinity boundaries and thermal shifts act as strict ecological barriers that select for specialized viral lineages.

Conclusion

This study unraveled the environmental drivers shaping giant virus biogeography across Chinese intertidal mudflats by coupling multivariate ordination with hierarchical classification. The results show that giant virus distribution is not random but structured by deterministic environmental sorting. A robust core cluster exists, featuring a large, resilient core estuarine community fueled by carbon and nutrient availability, surrounded by other clusters unique to a certain environment value. Thus, this research establishes a strong foundation for understanding the distributions of Nucleocytoviricota in coastal boundaries.

References

- Aylward, F. O., Moniruzzaman, M., Ha, A. D., & Koonin, E. V. (2021). A phylogenomic framework for charting the diversity and evolution of giant viruses. *PLoS Biology*, 19(10), e3001430–e3001430. <https://doi.org/10.1371/journal.pbio.3001430>
- Ha, A. D., Moniruzzaman, M., & Aylward, F. O. (2021). High transcriptional activity and diverse functional repertoires of hundreds of giant viruses in a coastal marine system. *MSystems*, 6(4). <https://doi.org/10.1128/msystems.00293-21>
- Ji, M., Li, Y., Wang, M., Liu, X., Gong, X., & Tu, Q. (2026). Unveiling the biodiversity of large DNA viruses in intertidal mudflats via metagenomics. *Nature Communications*, 17(1). <https://doi.org/10.1038/s41467-026-71095-7>

Appendix

Data Availability

The raw metagenomic dataset is available via DOI: 10.1038/s41467-026-71095-7, and the specific data and code used for these methods are deposited on GitHub (<https://github.com/mesosupp/NumEcolIndependentProject>).

Code Availability

```
# NumEcol Independent Project, Author: Devita Mayanda Heerlie,
Student ID: R14B44022

# -----
# PART 1: dbRDA with selected variables
# -----

# Load the library ----
library(vegan)
library(cluster)
library(dendextend)
set.seed(123) # For reproducibility

# Load data ----
otu                                     <-
read.delim('https://raw.githubusercontent.com/mesosupp/NumEcolI
ndependentProject/main/GVMAGs_OTU.txt', row.names = 1)
env                                     <-
read.csv('https://raw.githubusercontent.com/mesosupp/NumEcolInd
ependentProject/main/GVMAGs_env.csv', row.names = 1 )

# Data Preprocessing ----
env[env == "Na"] <- NA # Check the NA
env[] <- lapply(env, function(x) as.numeric(as.character(x)))
env <- env[, colMeans(is.na(env)) <= 0.20] # Drop columns with
>20% NA (TOC/Moisture)
env <- na.omit(env)

# Align and Log-Transform OTU Data ----
```

```

otu.t <- t(otu)
shared.samples <- intersect(rownames(otu.t), rownames(env))
env <- env[shared.samples, ]
otu.log <- log1p(otu.t[shared.samples, ])
otu.log <- otu.log[, colSums(otu.log) > 0] # Remove
zero-abundance OTUs

```

Calculate the Distance Matrix ----

```
bc.dist <- vegdist(otu.log, method = "bray")
```

Grouping and Visualization settings ----

```

OTU.group <- as.factor(gsub("[0-9]+$", "", sub("_.*", "",
rownames(env))))
pal <- c("#FF7F00", "#FC8D62", "#4DAF4A", "#A6D854", "#999999",
"#984EA3",
        "#E41A1C", "#990000", "#111BC3", "#377EB8", "#F781BF",
"#E78AC3",
        "#000000", "#FFFF00") # Colors for 14 Giant Virus OTUs
point.cols <- pal[OTU.group]

```

Function to set the margin and shows the legend

```

apply_legend <- function() {
  par(xpd = TRUE)
  legend(par("usr")[2] + 0.01 * diff(par("usr")[1:2]),
par("usr")[4],
        legend = levels(OTU.group), pch = 16, col =
pal[1:nlevels(OTU.group)],
        cex = 0.65, bty = "n", title = "OTUs")
  par(xpd = FALSE)}

```

Run The Analysis and Plot ----

Run the PCoA

```
pcoa <- cmdscale(bc.dist, eig = TRUE)
```

```
pcoa.eig <- round(pcoa$eig / sum(pcoa$eig[pcoa$eig > 0]) * 100,
1)
```

Plot 1: PCoA

```
par(mar = c(5, 5, 4, 10))
plot(pcoa$points[,1], pcoa$points[,2], pch = 16, col =
point.cols, cex = 1.3,
      xlab = paste0("PCoA1 (", pcoa.eig[1], "%)"), ylab =
paste0("PCoA2 (", pcoa.eig[2], "%)"),
      main = "PCoA (Bray-Curtis)")
apply_legend()
```

Envfit

```
env.fit <- envfit(pcoa$points, env, permutations = 999)
print(env.fit)
```

MAT, Salinity, pH, NO₂, TN, TP, SO₄, and TOC were highly significant (p<0.05)

Plot 2: PCoA + Environmental Vectors

```
par(mar = c(5, 5, 4, 10))
plot(pcoa$points[,1], pcoa$points[,2], pch = 16, col =
point.cols, cex = 1.3,
      xlab = paste0("PCoA1 (", pcoa.eig[1], "%)"), ylab =
paste0("PCoA2 (", pcoa.eig[2], "%)"),
      main = "PCoA + Environmental Vectors (p < 0.05)")
plot(env.fit, p.max = 0.05, col = "black", lwd = 2, cex = 0.85)
```

envfit drop SO₄

```
apply_legend()
```

Variable Selection & dbRDA

```
dbrda.all <- capscale(bc.dist ~ ., data = env)
dbrda.0 <- capscale(bc.dist ~ 1, data = env)
dbrda.sel <- ordiR2step(dbrda.0, scope = formula(dbrda.all),
direction = "forward",
                                R2scope =
RsquareAdj(dbrda.all)$adj.r.squared, permutations = 999)
```


ordiR2step appropriately dropped NH4 because they did not explain enough unique variation (p>0.10)

```
sel.vars <- attr(terms(dbrda.sel), "term.labels")
dbrda.final <- capscale(as.formula(paste("bc.dist ~",
paste(sel.vars, collapse = " + "))), data = env)
```

Plot 3: dbRDA with Selected Environmental Vectors

```
par(mar = c(5, 5, 4, 10))
ordiplot(dbrda.final, display = c("OTUss", "bp"), type = "n",
main = "dbRDA (Selected Variables)")
points(dbrda.final, display = "OTUss", pch = 16, col =
point.cols, cex = 1.2)
text(dbrda.final, display = "bp", col = "black", cex = 0.9,
arrow.mul = 1)
apply_legend()
```

-----

PART 2: Hierarchical Classification & Validation

-----

Cluster Color ----

```
clust.pal <- c("#1B9E77", "#D95F02", "#7570B3", "#E7298A")
```

Compute distances and Ward's clustering tree ----

Calculate the Ward's distance

```
bc.dist.ward <- sqrt(bc.dist)
clust.hclust <- as.hclust(agnes(bc.dist.ward, method = "ward"))
```

Define cluster memberships for both k=2 and k=4

```
groups_k2 <- cutree(clust.hclust, k = 2)
groups_k4 <- cutree(clust.hclust, k = 4)
```

Convert baseline tree to a dendrogram object

```
dend <- as.dendrogram(clust.hclust)
```

```
# Shared Helper Functions ----
```

```
## Create function to color the cluster
```

```
apply_cluster_legend <- function(k = 4) {  
  par(xpd = TRUE)  
  legend(par("usr")[2] + 0.01 * diff(par("usr")[1:2]),  
par("usr")[4],  
  legend = paste("Cluster", 1:k), pch = 16, col =  
clust.pal[1:k],  
  cex = 0.75, bty = "n", title = "Cluster")  
  par(xpd = FALSE)  
}
```

```
# Ward's Clustering Evaluation (k = 2 vs k = 4) ----
```

```
## k = 2 Tree
```

```
dend_k2 <- as.dendrogram(clust.hclust)  
labels_colors(dend_k2) <- point.cols[clust.hclust$order] #  
Insert the OTU color into the labels  
dend_k2 <- set(dend_k2, "labels_cex", 0.35)  
plot(dend_k2, main = "Ward's Clustering (k = 2)", xlab = "",  
sub = "") ## Plot k = 2 Tree  
cluster.order.k2 <- unique(groups_k2[clust.hclust$order])  
rect.dendrogram(dend_k2, k = 2, border = "red", lwd = 2) #  
Create the cluster
```

```
## k = 4 Tree
```

```
dend_k4 <- as.dendrogram(clust.hclust)  
labels_colors(dend_k4) <- point.cols[clust.hclust$order]  
dend_k4 <- set(dend_k4, "labels_cex", 0.35)  
plot(dend_k4, main = "Ward's Clustering (k = 4)", xlab = "",  
sub = "") # Plot k = 4 Tree  
cluster.order.k4 <- unique(groups_k4[clust.hclust$order])  
rect.dendrogram(dend_k4, k = 4, border =  
clust.pal[cluster.order.k4], lwd = 2)
```

PCoA Spatial Map + Ward Clusters Overlay ----

```
clust.cols <- clust.pal[groups_k4]
plot(pcoa$points[,1], pcoa$points[,2], pch = 16, col =
clust.cols, cex = 1.4,
      xlab = paste0("PCoA1 (", pcoa.eig[1], "%)"), ylab =
paste0("PCoA2 (", pcoa.eig[2], "%)"),
      main = "PCoA Space + Ward Clusters (k = 4)")
apply_cluster_legend(k = 4)
```

Alternative Linkage Method Validations ----

Single Linkage Tree

```
h.single <- hclust(bc.dist, "single")
dend.single <- as.dendrogram(h.single)
labels_colors(dend.single) <- point.cols[h.single$order]
dend.single <- set(dend.single, "labels_cex", 0.35)
plot(dend.single, main = "d. Single Linkage", xlab = "", sub =
"")
```

Complete Linkage Tree

```
h.complete <- hclust(bc.dist, "complete")
dend.complete <- as.dendrogram(h.complete)
labels_colors(dend.complete) <- point.cols[h.complete$order]
dend.complete <- set(dend.complete, "labels_cex", 0.35)
plot(dend.complete, main = "e. Complete Linkage", xlab = "",
sub = "")
```

Average Linkage Tree

```
h.average <- hclust(bc.dist, "average")
dend.average <- as.dendrogram(h.average)
labels_colors(dend.average) <- point.cols[h.average$order]
dend.average <- set(dend.average, "labels_cex", 0.35)
plot(dend.average, main = "f. Average Linkage", xlab = "", sub
= "")
```